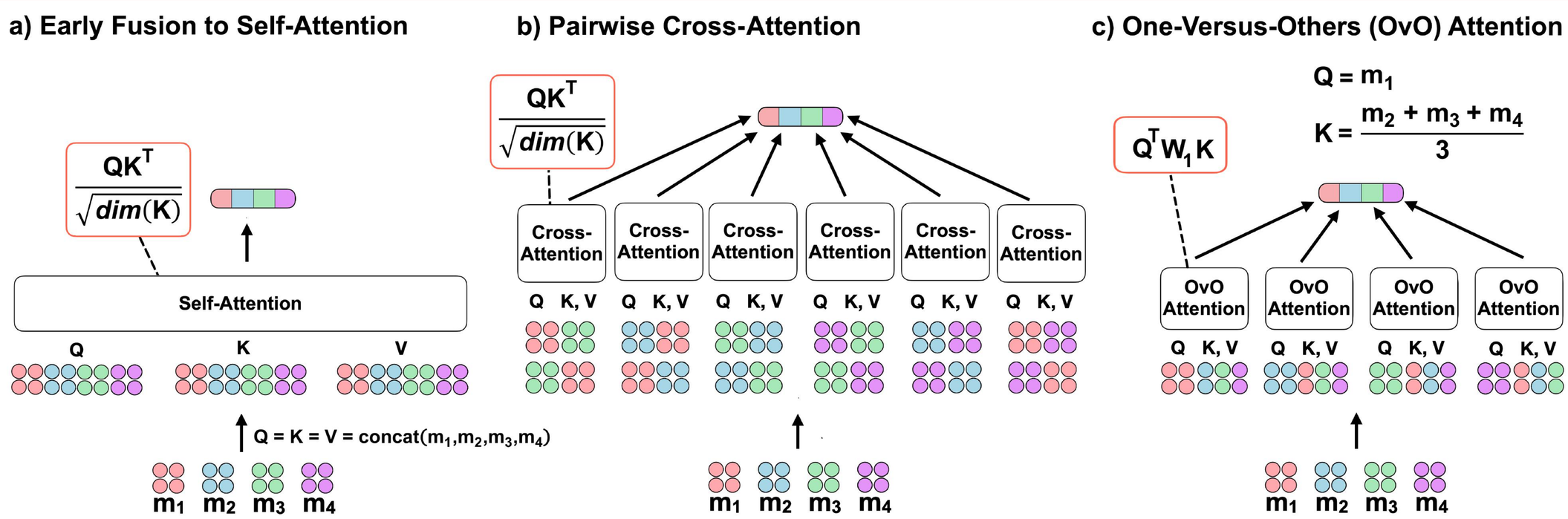
One-vs-Others Attention: Scalable Multimodal Learning

Michal Golovanevsky[1], Eva Schiller[1], Akira Nair[1], Eric Han[1], Ritambhara Singh[1,3], Carsten Eickhoff[1,2]

Brown University Department of Computer Science[1], Institute for Bioinformatics and Medical Informatics, University of Tübingen[2], Center for Computational Molecular Biology, Brown University[3]

Model Architecture



Overview

1. We propose a new attention mechanism, One-Versus-Others (OvO) attention, that **scales linearly** with the number of modalities
2. Using three biomedical datasets with diverse modalities, we show that our method decreases computation costs while increasing performance compared to popular integration techniques.
3. OvO reduced the number of required floating point operations (FLOPs) by at least 91.98%, demonstrating its significant impact on efficiency and enabling wider adaptation.

Results on Real-World Datasets

Table 2: MIMIC IV+CXR results. Modalities: Time Series (TS), Image (I), Demographics (D), and Text (T). We use AUROC and AUPRC following (Hayat et al., 2022).

Model	Modalities	↓ Δ FLOPs	↑ AUROC	↑ AUPRC
LSTM	TS	-	58.8 ±0.6	28.5 ±0.4
CNN	I	-	56.9 ±0.3	26.7 ±0.2
NN	D	-	64.1 ±0.4	32.4 ±0.3
ClinicalBERT	T	-	79.3 ±0.4	58.7 ±0.3
Concatenation	All	*	82.7 ±0.6	65.1 ±1.8
Cross-Attention	All	52,723,712	78.2 ±2.1	54.1 ±2.7
Self-Attention	All	67,633,152	78.5 ±2.0	55.7 ±3.1
OvO Attention	**All**	**4,227,072**	**83.6 ±1.1**	**66.2 ±2.6**

Table 3: TADPOLE results. Modalities: AV45 PET ROIs (A), CSF biomarkers (CSF), MRI ROIs (M), FDG PET ROI (F), Demographics (D), and Cognitive Tests (CT). We use multi-class area under the receiver operating curve (mAUC) and balanced classification accuracy (BCA), following (Marinescu et al., 2019).

Model	Modalities	↓ Δ FLOPs	↑ MAUC	↑ BCA
NN	A	-	63.5 ±3.1	56.4 ±3.8
NN	CSF	-	64.4 ±1.1	53.6 ±2.7
NN	M	-	67.0 ±1.3	57.2 ±1.0
NN	F	-	66.6 ±0.3	60.8 ±0.7
NN	D	-	74.6 ±0.9	62.0 ±0.6
NN	CT	-	97.8 ±0.2	88.6 ±0.7
Concatenation	All	*	97.7 ±0.8	91.9 ±1.9
Cross-Attention	All	8,921,088	97.1 ±0.6	90.7 ±1.7
Self-Attention	All	9,633,792	94.8 ±1.1	86.6 ±2.6
OvO Attention	**All**	**405,504**	**98.3 ±0.4**	**93.0 ±1.4**

Table 4: eICU results. Modalities: apacheApsVar (A), Demographics (DE), Medication (M), Lab (L), Diagnosis (D), and Treatment (T). We use AUROC and AUPRC following (Sheikhalishahi et al., 2020).

Model	Modalities	↓ Δ FLOPs	↑ AUROC	↑ AUPRC
NN	DE	-	50.2 ±0.6	91.8 ±0.2
NN	M	-	56.3 ±1.3	93.1 ±0.3
NN	D	-	58.2 ±2.1	93.3 ±0.4
NN	T	-	66.1 ±0.5	94.8 ±0.1
NN	A	-	77.6 ±0.2	97.0 ±0.1
NN	L	-	81.5 ±0.4	97.0 ±0.1
Concatenation	All	*	81.7 ±1.6	97.5 ±0.3
Cross-Attention	All	129,957,888	77.6 ±1.6	95.4 ±0.3
Self-Attention	All	151,781,376	80.2 ±2.0	96.8 ±0.4
OvO Attention	**All**	**6,340,608**	**82.5 ±0.9**	**97.8 ±0.2**

Complexity Analysis

Model	Complexity Per Layer
Self-Attention	$\mathcal{O}(k^2 \cdot n^2 \cdot d)$
Cross-Attention	$\mathcal{O}(k^2 \cdot n^2 \cdot d)$
One-Versus-Others (OvO) Attention	$\mathcal{O}(k \cdot n^2 \cdot d)$

Table 1: Complexity per layer for early fusion followed by self-attention, pairwise cross-attention, and our One-Versus-Others (OvO) attention. Let k represent the number of modalities, n the feature length of each modality, and d the representation dimension of the weight matrices. Self-attention and cross-attention have complexities of $O(k^2 \cdot n^2 \cdot d)$, while OvO attention scales linearly with k, resulting in a complexity of $O(k \cdot n^2 \cdot d)$.

Simulation Data Setting

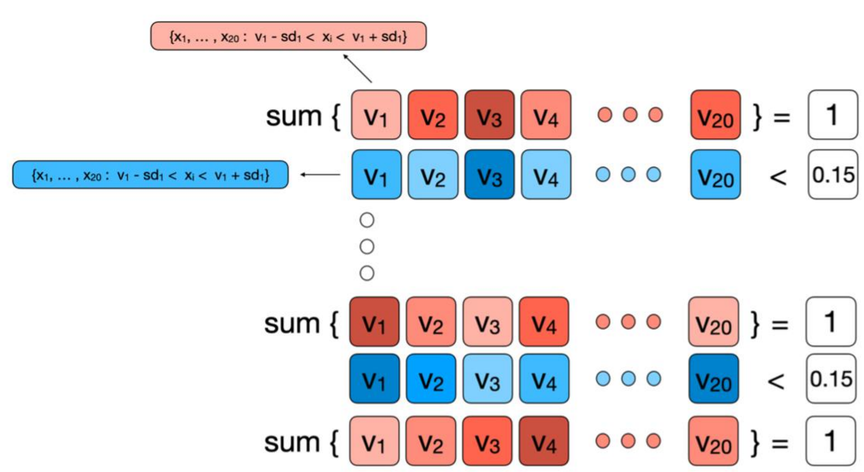


Figure 1: We simulated 20 modalities to test the capabilities of our method. The first classification label is created by 20 random values that add up to 1. Each value is then vectorized by sampling randomly around the chosen number. The second classification is created by randomly selecting numbers that are all under 0.15.

Simulation Data Results

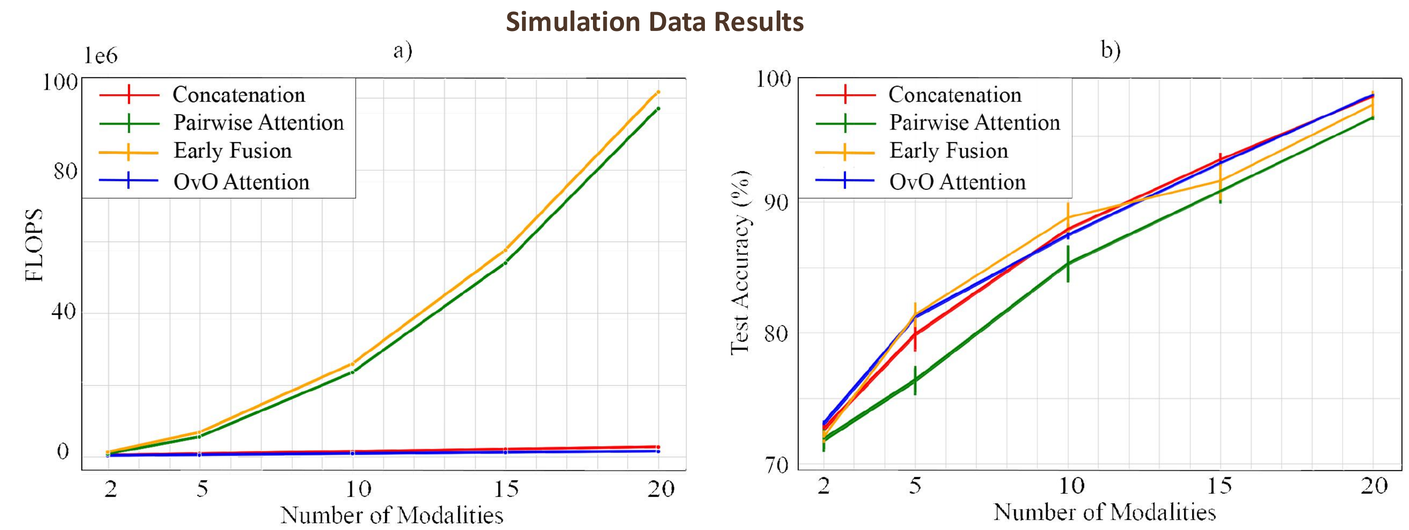


Figure 2: The impact of using OvO attention for fusion on compute (left) and performance (right) with increasing number of simulated modalities. OvO attention improves performance at lower computational cost.

One-vs-Others Attention

Attention takes as input queries and keys of dimension dk, and values. The attention score is computed as the scaled dot product:

$$Attention\ Score(Q,K,V) = \frac{QK^T}{\sqrt{d_k}}$$

To compute cross-modal attention, we can treat the query as inputs from one modality, and the keys and values as the inputs from another modality.
However, computing cross-attention between n modalities, results in $\binom{n}{2}$ number of calculations, or $2\binom{n}{2}$ if computing bi-directionally (e.g., vision to language, language to vision). Example is shown for 3 modalities:

$$a_{1,2} = Attention(Q = m_1, K = m_2, V = m_2)$$
$$a_{1,3} = Attention(Q = m_1, K = m_3, V = m_3)$$

$$a_1 = concat(a_{1,2}, a_{1,3})$$

We present **One-vs-Others attention**, which requires only as many calculations as there are modalities (n computations instead of $\binom{n}{2}$). One Versus Others Attention is as follows:

$$a_i = score(m_i, \{m_j : j \neq i\}) = m_i^T W_i \frac{\sum_{j \neq i}^{n} m_j}{n-1}$$

a_i is the attention score for the ith modality (m_i), n is the number of modalities, and W_i is a neural network weight matrix that scales all modalities.

The formula takes in one modality and compares it against all the other modalities with a weight matrix that can learn interactions throughout training (hence the name).

Email: Michal_golovanevsky@brown.edu
GitHub: https://github.com/rsinghlab/OvO